

Yitong Bai

M.S. Student in Computer Engineering, Columbia University

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🌐 brianbai

Research Interests

- LLM agents, multi-agent systems, ML systems, evaluation and verification, reproducibility, trustworthy AI, and LLM post-training.

Education

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|---|-----------------------|
| M.S. Columbia University , Computer Engineering | New York, NY |
| • Expected Dec 2026 | Sept 2025 – Dec 2026 |
| • GPA: 3.83 / 4.00 | |
| B.S. (Hons) Lancaster University , Computer Science | Lancaster, UK |
| • First Class Honours; GPA: 3.73 / 4.00 | Sept 2023 – June 2025 |
| • BJTU-Lancaster Dual-Degree Program (2+2) | |
| B.S. Beijing Jiaotong University , Computer Science and Technology | Beijing, China |
| • GPA: 3.37 / 4.00 | Sept 2021 – June 2023 |
| • BJTU-Lancaster Dual-Degree Program (2+2) | |

Research Experience

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| National Zhijiang Lab , Research Assistant | Hangzhou, China |
| Advisor: Jiayi Yang, Research Scientist, National Zhijiang Lab | June 2026 – present |
| <ul style="list-style-type: none">• Project: Feedback Learning for EHR SQL Repair: SFT vs. GRPO; manuscript in preparation.• Conducted a pre-registered study comparing SFT and GRPO on how diagnostic feedback shapes learning for EHR SQL repair, fine-tuning Qwen3.5-9B on MIMIC-IV EHR questions under four nested feedback levels, from a bare failure signal to repair hints.• Trained 12 QLoRA adapters across 4 feedback conditions x 3 seeds, and evaluated semantic repair success across 13 model groups and 10,400 records with task-clustered confidence intervals, permutation tests, and Holm correction.• Found no stable incremental benefit from richer feedback under SFT: matched gains of +1.0 to +2.0 percentage points did not meet the pre-registered significance bar.• Built a target-free GRPO pipeline with a custom binary execution reward and observed a non-monotonic feedback effect, with the intermediate L2 level reaching +10.0 points (95% CI [+4.0, +16.5], exploratory Holm-adjusted $p = 0.0228$), pending multi-seed confirmation. | |

Columbia University, CRIS Lab, Research Assistant

Advisor: Prof. Venkat Venkatasubramanian

New York, NY

Sept 2025 – present

- Project: DeepAudit: Agentic Evidence-Based Auditing of ML Reproducibility; manuscript in preparation.
- Designed and implemented DeepAudit, an agentic claim-level reproducibility auditing system that maps ML paper claims to executable repository tasks, metric contracts, runtime artifacts, and structured verdicts.
- Built a 23-step Python pipeline covering PDF-to-Markdown ingestion, table/figure-aware claim extraction, repository analysis, RAG-assisted code indexing, environment repair, autonomous execution, evidence alignment, scoring, and report generation.
- Formulated ML reproducibility auditing as claim-level evidence alignment rather than binary repository execution, distinguishing metric disagreement from missing data, incompatible environments, ambiguous entry points, unlogged metrics, and compute limits.
- Evaluated DeepAudit on 10 public ML paper-code pairs with 64 pre-specified audit targets; extracted 1,101 claims, identified 988 code-verifiable claims (89.7%), generated 49 metric contracts, and executed 52/64 experiment specifications (81.2%).

Columbia University / IBM Research Collaboration, Research Assistant

Mentored by Dhaval Patel, Senior Technical Staff Member, IBM Research

New York, NY

Feb 2026 – present

- Project: Multi-Turn AssetOps: Supervisor-Specialist Agents for Industrial O&M.
- Proposed and designed a LangGraph-based Supervisor-Specialist multi-agent architecture for multi-turn industrial asset operations and maintenance.
- Designed structured cross-turn artifact reuse for sensor evidence, anomaly analyses, failure hypotheses, and maintenance recommendations.
- Optimized routing, artifact reuse, and MCP execution with constraints, schema validation, duplicate-call suppression, and failure guards.
- Co-developed and evaluated 16 multi-turn industrial dialogs, reporting gains in planning effectiveness, task completion, tool-time share, and follow-up latency after artifact reuse.

SJTU Summer Campus, Project Leader

Facial Emotion Recognition with VGG16

Shanghai, China

July 2023 – Sept 2023

- Built a facial emotion recognition pipeline on the FER-2013 dataset using VGG16-based feature extraction and an MLP classifier.
- Applied data augmentation and feature fusion to improve robustness across emotion categories.
- Led a four-person team, coordinated task division and progress reviews, and completed both implementation and project reports.

Publications and Manuscripts

Towards Multi-Turn Dialog Systems for Industrial Asset Operations and Maintenance

2026

Chengrui Li, Rujing Li, *Yitong Bai*, Rui Liarxiv.org/abs/2605.24953 (arXiv preprint arXiv:2605.24953)**Application and Analysis of the VGG16 Model in Facial Emotion Recognition**

2023

Yitong Bai[10.5220/0012799800003885](https://doi.org/10.5220/0012799800003885) (Proceedings of DAML 2023, pp. 422-426)**Position: Agentic Science Needs Explicit Claim-Result-Computation Relationships**

2026

Yitong Bai

(Manuscript under review)

Beyond Static Leaderboards: Predictive Validity for the Evaluation of LLM Agents

2026

Dhaval Patel, *Yitong Bai*, et al.

(Manuscript under review)

The Label Is Not the Answer: Output-Schema Anchoring in Medical Multimodal MCQ Evaluation

2026

Chengyi Zhang, Yitong Bai, Ziyang Wang

(Accepted at MICCAI 2026 Challenge MedReason)

Industry Experience

Neusoft Group, Software Engineering Intern

Guangzhou, China

July 2024 – Aug 2024

- Contributed to development and testing for a university portal system, including frontend refinement, database interface support, and cross-database data matching.
- Worked with heterogeneous data sources to support synchronization and integration across subsystems.

Selected Projects

Agentic Java Formal Verification

Dec 2025

Columbia University, Formal Verification Project

- Built an agentic Java verification assistant that translates natural-language proof goals into Java annotations and iteratively repairs verifier errors.
- Designed a verification loop over OpenJML feedback, with prompt construction, error parsing, retry control, and proof-state tracking.

ConvNeXt and ViT for Hierarchical Open-Set Image Classification

Dec 2025

Columbia University, Neural Networks and Deep Learning Project

- Built a two-stage hierarchical open-set recognition pipeline in PyTorch for super-class and sub-class prediction.
- Implemented interchangeable ImageNet-pretrained ConvNeXt and ViT backbones for controlled comparison.
- Evaluated on a 6,288-image dataset with 3 super-classes and 87 sub-classes.
- Improved unseen sub-class recognition from 62.24% to 82.22% using ConvNeXt on a held-out novel split.

TrueShot

2026

Columbia University, Computer Networks Project

- Built a Dockerized Flask LAN game service with screenshot submission, peer validation, blockchain-style logging, and tamper-evident replay.
- Implemented and tested blockchain, networking, application, and web layers.

Computer Vision and ML Lead, Pet Companion Robot

Dec 2025

Columbia University, AIoT Project

- Led computer vision and machine learning components for an AIoT pet robot integrating on-device perception, motion control, cloud analysis, and dashboarding.
- Built real-time dog detection and motion control on Raspberry Pi 5 using YOLO, FastAPI, sensor fusion, and target tracking.
- Developed cloud pipelines for automated image capture, behavior summarization, and a VGG16-based pet emotion classifier.

Honors and Awards

- Honorable Mention, Mathematical Contest in Modeling, COMAP (Mar 2024)
- Beijing Jiaotong University Scholarship for Academic Progress, Top 5% (Jun 2022)

Skills

Programming: Python, C, Java, R, SQL, HTML

ML / LLM Tools: PyTorch, scikit-learn, NumPy, pandas, LangGraph, LangChain, RAG, MCP / FastMCP, LiteLLM, Pydantic, OpenAI API, IBM WatsonX

Systems / Engineering: Linux, Docker, Git, Conda / mamba / venv, pytest, Flask, FastAPI